MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10545707

(TIME DELAY)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of spceification MIL-C-50785 comprise the complete requirements for the Time Delay Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation forbids or request *for* proposal form a part of this specification to the extent specified herein,

SPECIFICATIONS

Military

MIL-C-50785	**Circuit Card Assemblies, General Specification for**

DRAWINGS

U.S. Army, Frankford Arsenal

10545425	**Sight Current Generator**
10545707	**Circuit Card Assembly, Time Delay**

PACKAGING DATA SHEET

10545707	**Packaging of Circuit Card Assembly Time Delay**

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the. contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Item definition. This assembly is a single ended printed circuit board consisting of an operational amplifier, a voltage limiting circuit, and a ten second time delay circuit. The circuit card is used to amplify, invert, and limit inputs which make up portions of the magnet and torque current voltages, also control the ten second time delay relay (K23) when installed in the Sight Current Generator, Drawing Number 10S4S42S.

3.1.1 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL) 10545707.

3.1.2 Government furnished property list. *Not* applicable.

3,2 Characteristics

3,2,1 Performance. The assembly shall, when assembled in accordance with Drawing Number 10545707, be capable of performing the following functions.

a, Operational amplifier Al shall amplify and invert one of the inputs required to develop the torque voltage.

b. Limit circuit Q1-Q4 shall limit the Vdc applied pin 5 to the value of Wc applied to pin 6. ,

c. Time delay circuit Q5, Q6 shall control relay K23 mounted in the sight current generator.

3,2.1.1 Input and output levels

3.2.1.1.1 Input power. The input power supplied shall be as specified in Table I under the conditions specified in Figure 1.

Table I. input Power

Pin	Voltage
4 and 25	+IS ± 2.0 percent Vdc
14	-15 ± 2.0 percent Vdc
1 and 28	Ground
5	+12 ± 2.0 percent

3.2.1.1.2 Output levels.

3.2.1.1.2.1 Pin 6 output. The voltage shall be within ± 0.100 Vdc of the voltage Vl, except it shall not exceed the voltage at pin 5, as VI varies from 1.0 Vdc to 20 Vdc.

3.2.1.1.2.2 Pin 10 output. The output shall be zero ± 50 millivolts between pin 10 and pin one with zero Vdc applied between pin 9 and pin one, and [Vin (-1.0)3 ± 0.5 percent with a Vin of -13 Vdc to +13 Vdc applied between pin 9 and pin one.

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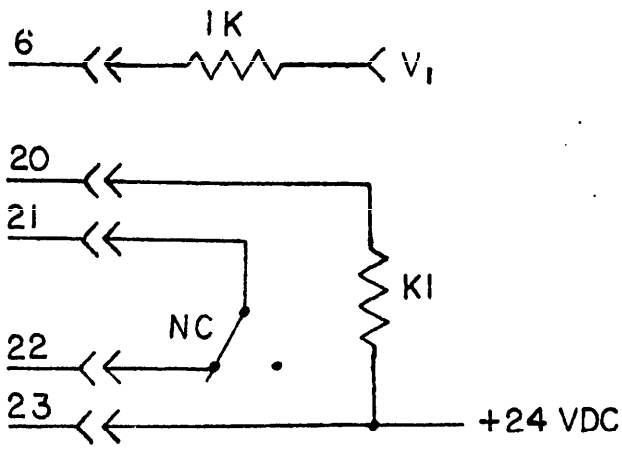


Figure 1. Circuit Diagram

3.2.1.1.2.3 Pin 20 output. The voltage between pin 20 and pin 28 shall drop to zero Vdc ± 1.0 Vdc from 24 Vdc in 10 ± 4.0 seconds after 24 Vdc is applied between pin 23 and pin 28.

3.2.1.2 Environmental. The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic. The physical characteristics shall be as specified in drawing 10545707 and the engineering parts list.

3.3 Design and construction

3.3.1 Production drawings. This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 10545707 and associated lists.

3.3.2 Standards of manufacture. The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship. The workmanship shall be as specified in specification MIL-C-50785.

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4. QUALITY ASSURANCE PROVISION

4.1 General

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination. The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment

4.2.1.1 Power supply (PS_1). The power supply shall be 15 ± 0.1 Vdc, 100 milliamperes or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS_2). The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes or greater, ripple less than 10 mv rms or 100 mv peak-to-peak.

4.2.1.3 Power supply (PS_3). The power supply shall be zero ± 13 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 10 ma or greater.

4.2.1.4 Power supply (PS_4). The power supply shall be 24 ± 0.5 Vdc, 100 ma or greater.

One resistor is required with a rating of 4000 ohms ±1.0 percent, 1/4 watt.

± 1.0 percent, 1/ watt. One resistor is required with a rating of 16,000 ohms

± 5.0 percent, 1/2 watt. One resistor is required with a rating of 1000 ohms

4.2.1.8 Relay (K1). One relay, MS25329 - D1 or equivalent, is required.

4.2.1.9 Switch (S1). One single pole, single throw switch is required.

4.2.1.10 Digital voltmeter. The digital voltmeter (DVM) shall have a voltage accuracy and resolution of 0.01 percent or better from zero to 100 Vdc. Input impedance greater than 10 megohms on all voltage scales.

4.2.1.11 Oscilloscope. The oscilloscope shall have voltage, time accuracy and resolution of 5.0 percent or better, with a rise time of 100 nanoseconds or better and with external triggering. Voltage sensitivity shall include the range from 0.05 volts/cm to 5.0 volts/cm.

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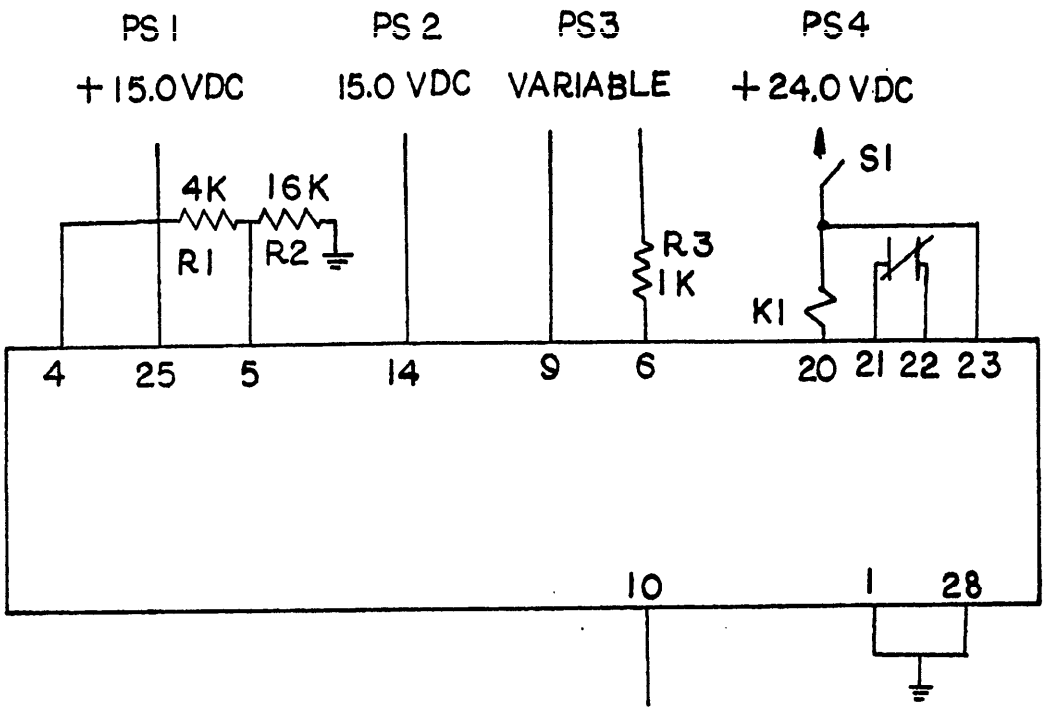


Figure 2. Test Connections

4.2.2 Functional tests

4.2.2.1 Test connections. The power supplies, resistors, and relay shall be connected to the assembly under test in accordance with Figure 2 using receptacle 10545455.

4.2.2.2 Test procedure. Compliance with the requirements of 3.2.1.1.1 through 3.2.1.1.2.3 shall be verified as follows:

a. Power supply (PS_3) voltage shall be adjusted and connected to the specified input pin per Table II. A digital voltmeter shall be used to measure the voltage at the applicable output pin to the limits indicated in Table II.

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b. The voltage at pin 20, as measured on the oscilloscope, shall be zero ± 1.0 Vdc, rising to +24 ± 2 Vdc when switch S1 is closed, then returning to zero ± 1.0 Vdc after a delay of 10 ± 4 seconds.

Table II. Test Limits

Input Voltage (Vdc)	Input Pin	Output Voltage (Vdc)	Output pin
+1,0 ± 0.010	6 (thru lK]	+1.0 ± 0.110	6
+10.0 ± 0.010	6 (thru lK]	+10.0 ± 0.110	6
+20.0 ± 0.020	6 (thru lK)	Less than pin 5 voltage	6
+13.0 ± 0.013	9	-1300 ± 0.06S	10
() ± ().()10	9	o ± 0.050	10
-1300 ± 0.013	9	+13.0 ± 0.065	10

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC form 1029, 10545707.

6. NOTES

6.1 The notes specified in the general specification MIL-C-S0785 are applicable to this specification.

Custodian
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Preparing Activity
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Project No. 1285-A317

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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